

Valdoria Votes: A Political Mystery (Part 2)

Question Menu

1. Investigate Alerts
2. Analyze Logs
3. Write Reports

<https://kc7cyber.com/challenges/335>

Your mission: investigate the events leading up to this critical election. Analyze the data, uncover any hidden agendas, and help protect the integrity of Valdoria's vote.

Tool: KQL (Kusto Query Language)

What is the name of the Deputy Commissioner?

Employees

```
| where role == "Deputy Commissioner"
```

Name: Hilary Binton

What is Dora Thomas' role?

Employees

```
| where name == "Dora Thomas"
```

Polling Station Supervisor

What is this supervisor's name? Temp Election Support Staff Supervisor

Employees

```
| where role contains "supervisor"
```

Barry Schmelly

What is Barry Schmelly's IP address?

Employees

```
| where name == "Barry Schmelly"
```

10.10.0.12

What is Barry Schmelly's hostname?

Hostname= GCH3-DESKTOP

Email address

barry_schmelly@valdoriavotes.gov

How many emails did Barry Schmelly receive?

Email

```
| where recipient == "barry_schmelly@valdoriavotes.gov"
```

```
| count
```

37

How many distinct commands were run on Barry Schmelly's machine?

ProcessEvents

```
| where hostname == "GCH3-DESKTOP"
```

```
| distinct process_commandline
```

173

How many distinct URLs did employees with the first name William visit?

```
let wills_ips =
```

Employees

```
| where name has "William"
```

```
| distinct ip_addr;
```

```
OutboundNetworkEvents
| where src_ip in (wills_ips)
| distinct url
217
```

How many authentication attempts did we see to the accounts of employees with the first name William?

```
let w_username = Employees
|where name has "William"
|distinct username;
AuthenticationEvents
| where username in (w_username)
183
```

When in doubt you can do a `take 10` to quickly peek into

```
let w_username = Employees
|where name has "William"
|distinct username;
AuthenticationEvents
| where username in (w_username)
| take 10
```

Which table would tell us if Barry Schmelly received a suspicious email?

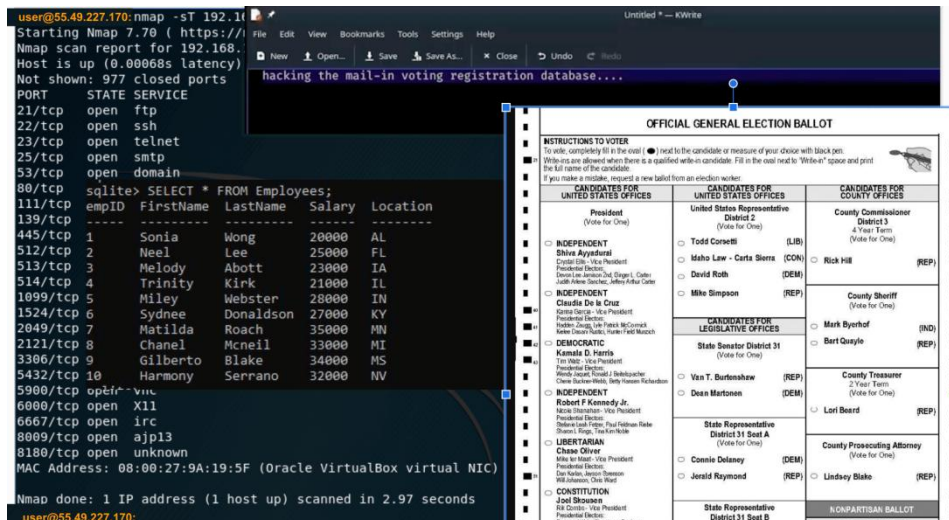
Table Name	Description	Example
 AuthenticationEvents	Logins to devices (successes and failures).	When someone tries to log in to their email account.
 Email	Emails sent and received by employees.	When an employee sends or receives a work email.
 Employees	Info about the company's employees.	Names, job titles, and contact details of employees.
 FileCreationEvents	Files created on employee devices.	When someone saves a new document on their computer.
 InboundNetworkEvents	Internet activity coming into the company network.	When someone visits the company website.
 OutboundNetworkEvents	Internet activity going out from the company network.	When someone visits an external website like google.com.
 PassiveDns (External)	Which IP addresses match up with which websites.	Figuring out that "google.com" matches with IP "8.8.8.8".
 ProcessEvents	Programs that start running on employee devices.	When an employee opens a program like Microsoft Word.
 SecurityAlerts	Security warnings from devices or email systems.	When antivirus software detects a virus on a computer.

Table very useful to be translate to other sequel language

An unknown hacking group claims to have compromised the voting machines in Valdoria. But how? The voting machines are supposed to be extremely hard to hack! The voting systems are airgapped. So how did they actually do it? How could they have done it?

What IP address did they forget to hide in the poster?

55.49.227.170



Is there any evidence of traffic from this IP address to our network? (yes/no)

No

We need a ping, only shown are open port with the tool Nmap

How many domains resolved to the IP 55.49.***.***?

PassiveDns

| where ip contains "55.49."

55.49.227.170

Domains found

valdoriavotesgov.com

shadow-hackers-r.us

What is the Valdoria related domain resolving to that IP?

valdoriavotesgov.com

What is the legitimate domain for the Valdoria Elections Board?

valdoriavotes.gov

How many IP addresses did the fraudulent Valdoria government domain resolve to?

PassiveDns

| where domain contains "valdoriavotesgov.com"

214.85.104.248

55.49.227.170

157.100.244.104

Which table should we be looking at for records of that kind of activity again?

InboundNetworkEvents any browsing activity from these IPs to our network

How many requests do we see to our network from those IPs?

let ips = PassiveDns

| where domain == "valdoriavotesgov.com"

| distinct ip;

InboundNetworkEvents

| where src_ip in (ips)

26

Which phase of the Cyber Kill Chain is this called?

recon;reconnaissance

let ips = PassiveDns

```
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "prevention"
election interference
```

What important machines were the hackers trying to get into? (two words)

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "machine"
voting machines
```

Which document were the hackers looking for? (two words)

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
| where url contains "document"
```

Which document were the hackers looking for? (two words)

```
let ips = PassiveDns
| where domain == "valdoriavotesgov.com"
| distinct ip;
InboundNetworkEvents
| where src_ip in (ips)
technical manual
```

Let's check if there's any traffic to it—has any of our employees visited that domain for any reason? (yes/no)

```
OutboundNetworkEvents
| where url contains "valdoriavotesgov.com"
```

Yes 10.10.0.4

```
Employees
| where ip_addr == "10.10.0.4"
```

Anderson Snooper

When did someone first browse to that domain?(paste the full timestamp)

2024-10-07T10:46:45Z

Did anyone enter their credentials on that phishing page ? (yes/no)

Yes

2024-10-07T10:46:47.000Z

What is the username of the employee that entered their credentials on that phishing page?

```
Employees
| where ip_addr == "10.10.0.4"
```

ansnooper

Anderson Snooper

Temp Election Support Staff Lead

False

What table would we find this logon activity in?

AuthenticationEvents

Table Name	Description	Example
 AuthenticationEvents	Logins to devices (successes and failures).	When someone tries to log in to their email account.
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When did the threat actor login to Snooper's account?

From the previous question we know that `valdoriavotesgov.com` as 3 SRC_IP

`214.85.104.248`

`55.49.227.170`

`157.100.244.104`

So we can write the SQL

`AuthenticationEvents`

```
| where username == "ansnooper"
| where timestamp between(datetime(2024-10-05T10:46:47Z) .. datetime(2024-10-11T10:46:47Z))
| where src_ip contains "214.85.104.248" or src_ip contains "55.49.227.170" or src_ip contains "157.100.244.104"
```

We found `157.100.244.104` was reached

`2024-10-07T15:46:45.000Z`

What is Anderson Snooper's email address?

`Employees`

```
| where ip_addr == "10.10.0.4"
anderson_snooper@valdoriavotes.gov
```

On October 8th, 'Snooper' had an interesting email conversation that's worth investigating

What is the email address of the person he was conversing with?

`Email`

```
| where recipient == "anderson_snooper@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
barry_schmelly@valdoriavotes.gov
```

What is Barry Schmelly's IP address?

`Employees`

```
| where name == "Barry Schmelly"
```

What is Schmelly's job role?

Temp Election Support Staff Supervisor

"Snooper" was observed asking Schmelly how one might gain access to what devices?

Email

```
| where recipient == "anderson_snooper@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
| where subject contains "machine"
voting machines
```

Email

```
| where sender == "anderson_snooper@valdoriavotes.gov"
| where recipient == "barry_schmelly@valdoriavotes.gov"
| where timestamp between ( datetime(2024-10-08)..datetime(2024-10-09))
do my job though
```

AI system

What is Snooper's IP address?

OutboundNetworkEvents

```
| where url contains "valdoriavotesgov.com"
10.10.0.4
```

Employees

```
| where ip_addr == "10.10.0.4"
```

ansnooper

Anderson Snooper

Temp Election Support Staff Lead

What term appeared at the end of each url that Snooper guessed?

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
| distinct url
```

model=gpt-4o

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
```

Select early time

2024-10-16T15:57:05.000Z

Response= Ai

What status code was returned for Snooper's unsuccessful guesses?

Response= 404

What was the nursery rhyme-related subdomain that Snooper guessed?

ol-mcdonald-had-a-farm-ai-ai-oh

What was the basketball player-related subdomain that Snooper guessed?

allen-iverson

What was the first subdomain "Snooper" guessed that returned a 200 status code?

https://elections-chatbot.valdoriavotes.gov/?model=gpt-4o

elections-chatbot

How many questions did they ask the chatbot?

```
let Snooper_ips =Employees
| where name has "Snooper"
| distinct ip_addr;
InboundNetworkEvents
| where src_ip in (Snooper_ips)
| where url contains "elections-chatbot.valdoriavotes.gov/?model=gpt-4o?prompt"
```

Response=6

By checking the AIPrompts table, we can find out what response the AI chatbot gave.

Which conversation_id is associated with the question about voting machines?

AIPrompts

```
| where prompt has "voting machines"
| project conversation_id
```

94bd6162-1323-402d-bccd-8fceaee5f230

Let's "pivot" on that conversation_id to see what else that user searched for.

The AI bot told the threat actor to bring a _ and a banana.

AIPrompts

```
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "banana"
```

bring a flashlight and a banana

flashlight

According to the bot, the voting machines are not actually connected to the internet.

Instead, votes are manually calculated using a ____.

AIPrompts

```
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "calcul"
Calculator
```

After trolling the threat actor for a bit, the AI chatbot suggests they might have better luck getting information about the voting machines from the vendor who makes them.

What is the name of the vendor?

AIPrompts

```
| where conversation_id == "94bd6162-1323-402d-bccd-8fceaee5f230"
| where response contains "vendor"
```

Voting vendor= Dominos Voting Systems
Election Commissioner

And who is that?

What is the name of the employee with that job role?

```
Employees  
| where role == "Election Commissioner"
```

Name = Arrack Bobama
Username= arbobama

Following the password reset, the threat actors logged into Bobama's account.

When did they log in to Bobama's account?

```
let w_username = Employees  
| where name has "Bobama"  
| distinct username;  
AuthenticationEvents  
| where username in (w_username)  
| where src_ip contains "214.85.104.248" or src_ip contains "55.49.227.170" or src_ip  
contains "157.100.244.104"  
| take 10
```

2024-10-16T00:00:00.000Z

What IP address did the threat actors use to log in to that account?

Src_ip= 214.85.104.248

The threat actors (using Bobama's account) were later observed sending an email to the voting machine vendor.

What email address did they send this email to?

```
Employees  
| where role == "Election Commissioner"
```

Ip_addr= 10.10.0.13

Email: arrack_bobama@valdoriavotes.gov

Voting vendor= Dominos Voting Systems

```
Email  
| where sender == "arrack_bobama@valdoriavotes.gov"  
| where recipient contains "DominosVotingSystems"  
| where timestamp > datetime(2024-10-16)
```

help@dominosvotingsystems.com

In this conversation, the threat actors are once again informed that the voting machines can only be accessed physically, as they are not connected to the internet.

What PDF did the threat actors receive that might be useful to them later?

```
Email  
| where recipient contains "arrack_bobama@valdoriavotes.gov"  
| where attachments contains "pdf"  
| where timestamp > datetime(2024-10-16)
```

ValdoriaVotingMachinesNetworkGuide.pdf

What You'll Learn

deal with yet another election drama in Valdoria 

Core Analyst Competencies

Choosing the Right Data Source

Documenting Your Work

Recognizing Investigation Starting Points

Querying Logs

Pivoting Across Data

Catching Your Own Blind Spots

Forming and Testing Hypotheses

Spotting Abnormal Activity

Domain-Specific Knowledge

Mapping Attacker TTPs

Recognizing Common Entry Points

Understanding Malicious Email Content

Understanding How SIEMs Work

Understanding Active Directory

Reading Process Trees

Advanced Topics

Handling Cloud Forensics Challenges